PROPRIETARY & CONFIDENTIAL — LWANDILE ENGINEERING PROJECTS & INCLUSIVE ENTREPRENEURIAL ACADEMY

SCAPUSH PRECISION STANDARD

SECTION 06 — IDEAL MVP SPECIFICATION & BUDGET REGISTER

Sovereign Bio-Solar-Thermal Campus | JET EOI Locked Baseline

Doc Ref: 06_mvp_scaled_budget_register.html | Status: LOCKED SCALED MVP BASELINE | Governance: Scapush Precision | Total Budget: R 3,734,500.00

Auto-Scrolling Live Narration Engine

▶ Start Auto-Scroll Narration ■ Stop / Misa

1. Executive Strategy & Locked Scaled Capacity Baseline

LISTEN SECTION

This specification locks in the definitive Ideal Minimum Viable Product (MVP) baseline for the Sovereign Bio-Solar-Thermal Campus. The scope features fully integrated parameters: **20,000m² leasehold perimeter boundary enclosed by a heavy-duty 2.2-meter high security fence**, **4,800 Broilers across 4 Pods (12m × 32m Barn 1)**, **160 Swine Weaners across 4 Pods / 8 Pens (12m × 32m Barn 2)**, a **70m³ Anaerobic Digester**, a **30 kWp Solar PV Array**, a **15 kW Biogas CHP Engine**, and a **60 kWh LFP BESS Bank**.

LOCKED MID-SCALE PROOF-OF-CONCEPT MANDATE

By fully incorporating the 20,000m² secure perimeter boundary and commercial-scale biological production units, the campus achieves complete sovereign biosecurity and continuous 15 kW biogas CHP baseload operation. All baseline comparisons are finalized; this scaled architecture represents the official lock-in for JET EOI submission and project execution.

2. Chronological Build Phases & Technical Specifications

LISTEN SECTION

Chronological Execution Roadmap (Phases 01 to 06)

```
[ Phase 01: 20k m² Perimeter Fence & Water Civil ] → [ Phase 02: 30kWp Solar / 60kWh BESS ]
                                                  ↓
[ Phase 03: Barn 1 (4,800 Broilers) & Barn 2 (160 Swine) ] → [ Phase 04: 70m³ Digester & 15kW CHP ]
                                                  ↓
[ Phase 05: Climate Nursery Pod & Central SCADA ] → [ Phase 06: Zone M Accessibility & Academy ]
```

Build Phase	Chronological Infrastructure Package	Locked MVP Technical Specifications & Scope	SCAPUSH Ref
Phase 01	20,000m² Perimeter Fence & Water Sovereignty	**• 20,000m² Site Perimeter Security Fence (2.2m height, diamond mesh / electric strand topping)** • Borehole-01 drilling (80m–100m depth, 165mm uPVC casing) • 1.5kW Brushless DC Solar Submersible Pump (75m–90m TDH) • 2 × 10,000L Elevated Polyethylene Tanks on 6m Steel Tower (20,000L Total) • Multi-stage filtration (Media sand + 5-micron + UV sterilizer)	**MVP-PH01-R2**

Build Phase	Chronological Infrastructure Package	Locked MVP Technical Specifications & Scope	SCAPUSH Ref
Phase 02	30 kWp Solar PV & 60 kWh BESS Microgrid	• 30 kWp Solar PV Array (Ground-mount tilt frames + rooftop integration) • 60 kWh Usable LFP (LiFePO4) BESS Battery Bank (Zone L Enclosure) • 30 kVA Off-Grid Multi-Mode Hybrid Storage Inverter / Grid-Former • Central AC Distribution Board + Cable Reticulation to Barns & Energy Centre	**MVP-PH02-R2**
Phase 03	Barn 1 & Barn 2 Livestock Infrastructure	• 10-Layer Biosecurity Station (Shower-in/out + Vehicle spray arch) • Barn 1 (12m × 32m = 384m^2): 4 Pods holding 4,800 Broilers total • Barn 2 (12m × 32m = 384m^2): 4 Pods / 8 Pens holding 160 Swine Weaners • 2% Sloped concrete floors + automated gravity slurry flumes	**MVP-PH03-R2**
Phase 04	70m^3 Biodigester & 15 kW CHP Engine	• 70m^3 Fixed-Dome Anaerobic Biodigester Vessel (Mesophilic 37°C) • Multi-stage Gas Scrubber Unit (H2S removal + Moisture condensate trap) • 15 kW Biogas CHP Generator Unit • Exhaust & Jacket Hydronic Water Heat Exchangers for underfloor climate loops	**MVP-PH04-R2**
Phase 05	Climate Controlled Nursery & SCADA Hub	• Modular Climate Nursery Pod (Zone F: Dual hydronic floor loops) • SCADA Control Room Terminal (Zone A: PLC + Dual Monitor HMI) • 120 Active Telemetry Tags (Temp, Pressure, Level, SOC, Flow, Gas) • Priority 1 Alarm Siren + GSM SMS Dialer module	**MVP-PH05-R2**
Phase 06	Accessibility Hub & Academy Lab	• Zone M Step-Free Accessible Pathway Corridor (1:15 max ramp) • 4 × 24V/48V Wheelchair Charging Bays (5.0 kWh/day allocation) • 30 kW DC Fast EV Charger + 7.4 kW Single-Phase AC Charger • Modular Academy Container Classroom (30 Trainee capacity)	**MVP-PH06-R2**

3. Locked Itemized Budget Register with 10% Contingency (ZAR / RAND)

LISTEN SECTION

The following detailed cost matrix reflects the definitive, locked CAPEX baseline for the Sovereign Bio-Solar-Thermal Campus, incorporating the 20,000m^2 security fence and all commercial production subsystems:

Ref Code	Budget Item Category	Specific Locked MVP Hardware & Works Scope	Estimated Cost (ZAR)
BUD-001A	20,000m^2 Perimeter Security Fence	**2.2m high heavy-duty security fence (approx. 570m perimeter run, security gates & posts)**	**R 185,000.00**
BUD-001B	Water Infrastructure & Civil Earthworks	Borehole drilling, casing, 1.5kW DC pump, 2x10kL tanks, 6m steel tower, UV plant, site leveling	**R 245,000.00**
BUD-002	30 kWp Solar PV System	30 kWp Tier-1 Monocrystalline panels, ground/roof mounting frames, DC combiners, cabling	**R 310,000.00**
BUD-003	60 kWh BESS & Inverters	60 kWh Tier-1 LFP battery bank, 30 kVA hybrid storage inverter, DC protection switchgear	**R 360,000.00**
BUD-004	70m^3 Biodigester Plant	70m^3 underground digester masonry/liner, gas holder, bio-scrubbers, condensate traps	**R 265,000.00**
BUD-005	15 kW Biogas CHP Engine	15 kW heavy-duty biogas generator, thermal heat recovery exchangers, circulation pumps	**R 225,000.00**
BUD-006	Barn 1: Poultry (4,800 Chicks)	12m × 32m Barn 1 structure (4 Pods), drinker/feeder lines, curtains, 2% sloped concrete slab	**R 480,000.00**
BUD-007	Barn 2: Swine (160 Weaners)	12m × 32m Barn 2 structure (4 Pods / 8 Pens), drinkers, gravity slurry channels, pen gating	**R 450,000.00**

Ref Code	Budget Item Category	Specific Locked MVP Hardware & Works Scope	Estimated Cost (ZAR)
BUD-008	Nursery & Thermal Loop	Modular insulated nursery pod, hydronic underfloor PEX piping, distribution manifolds	**R 145,000.00**
BUD-009	Biosecurity & Sanitation Station	Shower-in/out container, automated vehicle spray arch, chemical dosing, PPE lockers	**R 135,000.00**
BUD-010	SCADA, Sensors & Controls	Industrial PLC, dual HMI monitors, 120 telemetry sensors, GSM dialer, control desk	**R 120,000.00**
BUD-011	Accessibility & EV Hub	Step-free paving, 4x wheelchair charging bays, 30kW DC Fast EV charger + AC charger	**R 185,000.00**
BUD-012	Academy Modular Lab	12m insulated container classroom, IT desks, seating, AV system (30 trainee capacity)	**R 110,000.00**
BUD-013	Engineering & Oversight	Scapush precision system integration, technical oversight, commissioning & testing	**R 180,000.00**
		LOCKED SUBTOTAL MVP CAPEX:	**R 3,395,000.00**
		10% MANDATORY CONTINGENCY:	**R 339,500.00**
		TOTAL LOCKED SCALED MVP CAPEX (WITH CONTINGENCY):	**R 3,734,500.00**

4. Locked Technical Architecture Summary

LISTEN SECTION

Comprehensive Infrastructure Parameters

Infrastructure Parameter	Locked Baseline Specification	Strategic Operational Function
Perimeter Boundary Security	**20,000m² 2.2m High Security Fence**	Full sovereign boundary lockdown and biosecurity integrity.
Poultry Production (Barn 1)	**4,800 Chicks (4 Pods / 384m²)**	Full commercial batch scale generating primary protein revenue.
Swine Production (Barn 2)	**160 Swine (8 Pens / 384m²)**	Full barn utilization; yields optimal continuous slurry volume.
Anaerobic Biodigester	**70m³ Fixed-Dome Digester**	Sustains continuous gas pressure for 24/7 CHP operation.
Solar PV & BESS	**30 kWp Solar / 60 kWh LFP BESS**	Robust off-grid generation and storage buffering for campus loads.
Biogas CHP Engine	**15 kW Biogas CHP Engine**	Delivers 100% continuous hydronic heating for climate loops.

5. Locked MVP Master Register & JET EOI Declaration

LISTEN SECTION

Ref Code	Locked MVP Parameter	Governance & EOI Submission Basis
MVP-S01-F	20,000m² Perimeter Security Fence	Heavy-duty 2.2m high boundary fence enclosing the complete leasehold footprint.
MVP-S02	4,800 Broilers / 160 Swine	Full Barn 1 and Barn 2 spatial utilization (2 × 384m² footprints).
MVP-S03	70m³ Digester / 15kW CHP	High-volume bio-energy engine driving 100% hydronic thermal balance.
MVP-S04	30 kWp Solar / 60 kWh BESS	Balanced off-grid energy generation and fast-acting storage buffering.

Ref Code	Locked MVP Parameter	Governance & EOI Submission Basis
MVP-S05	20,000L Water Autonomy	Borehole-01 supplying 2x10kL elevated tanks giving 3.0 days safety reserve.
MVP-S06	120-Tag SCADA Platform	Comprehensive telemetry tracking all livestock, energy, and water loops.
MVP-S07-F	Locked Total Budget Baseline	ZAR 3,734,500.00 inclusive of 20,000m² fence and 10% contingency.

6. Approval & Authorization

LISTEN SECTION

DOCUMENT STATUS

STATUS: OFFICIALLY LOCKED BASELINE (REV 3.0) — 20,000M² FENCED & SCALED MVP BUDGET (ZAR 3,734,500.00 LOCKED FOR JET EOI)

PROJECT STEWARD & TECHNICAL LEAD



Select File | Lock Signature

Sipho Wiseman Khuzwayo
Inclusive Legacy Architect / Principal Lead
Lwandile Engineering Projects (Pty) Ltd
Status: Signature Image Loaded

GOVERNANCE & ACADEMIC LEAD



Select File | Lock Signature

Dr. Palesa Charlotte Felix-Faure
Founder & Academic Lead
Inclusive Entrepreneurial Academy
Status: Signature Image Loaded

Consortium Leadership
Sipho Wiseman Khuzwayo — Lead System Architect (Lwandile Engineering)
Dr. Palesa Charlotte Felix-Faure — Academic & Governance Lead (Inclusive Entrepreneurial Academy)

Execution Entities & Standard
Lwandile Engineering Projects (Pty) Ltd
Inclusive Entrepreneurial Academy
Scapush Precision Execution Standard

Operational Footprint
Sovereign Bio-Solar-Thermal Campus, Mpumalanga, South Africa

Primary Contact & Communication
siphokhuzwayo12@gmail.com
LinkedIn Profile